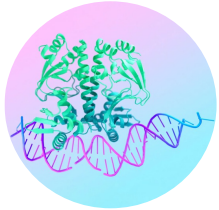
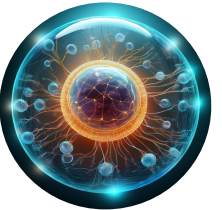


Current state of AI agents in life sciences



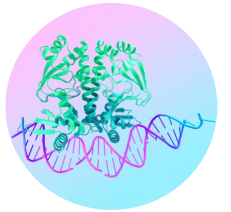
Mahbub Ul Alam
Data Engineer (AI)
SciLifeLab Data Centre
mahbub.ul.alam@scilifelab.se



By the end of this webinar, **you will...**



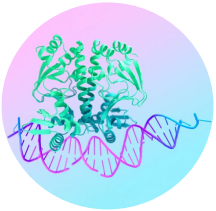
1. **Understand** what AI agents are and why they matter for research
2. **Know** which agent to choose for your specific workflow
3. **Be aware of** best practices for responsible use
4. **Have a glimpse** into the future of AI agents in life sciences
5. (🎁 **Bonus!**) **Explore an interactive AI agent landscape tool** 😊



Outline



1. LLMs vs. AI agents: a critical distinction



2. Current AI agents in life science workflows



3. How to get started with these tools



4. Emerging trends & what's next

From raw LLMs to AI agents: a comparison



Feature	Raw LLM	Chat interface	AI agent
What it is	Base model that predicts next tokens	LLM wrapped with UI, memory, and safety layers	LLM-powered system that autonomously pursues goals
Example	GPT-4 base	ChatGPT, Gemini	We'll see soon 😊
Data privacy & security	Data sent to external provider; limited control over sensitive information	Provider-controlled; data may be logged, used for training, or stored externally	Can run locally or on-premise; full control over data flow, tool permissions, and which systems access sensitive information

From raw LLMs to AI agents: a comparison



Feature	Raw LLM	Chat interface	AI agent
Interaction mode	Single input → single output	Multi-turn conversations with context retention	Goal-directed, iterative task execution
Memory	Stateless between calls	Session-based context window	Can maintain short and long-term memory across tasks
Tool access	None	Often includes search, file upload, code execution	Extensive, APIs, databases, browsers, external systems

From raw LLMs to AI agents: a comparison



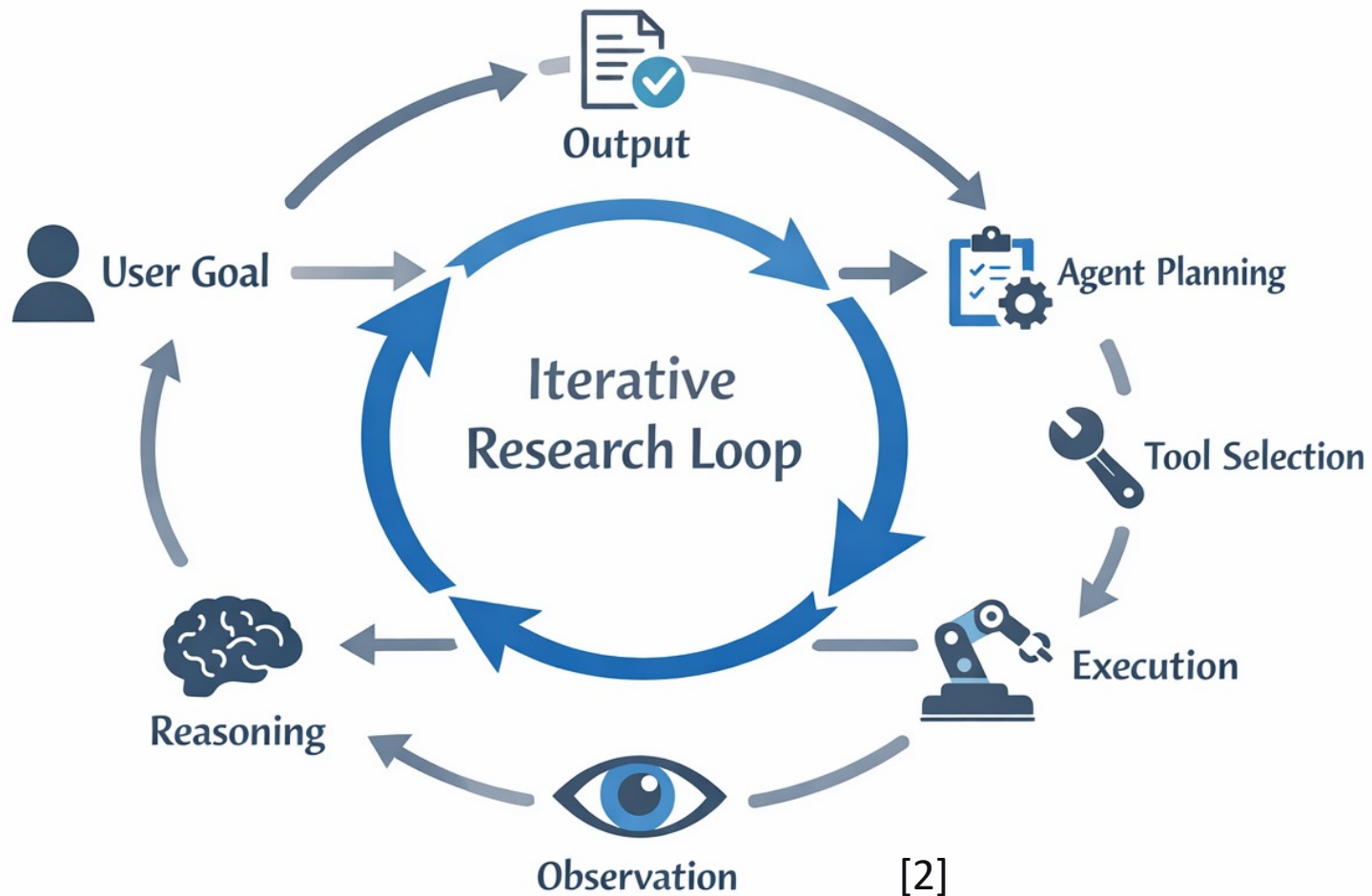
Feature	Raw LLM	Chat interface	AI agent
Decision making	None	Limited to response generation	Breaks down goals, selects tools, handles errors, iterates
Best for	Embedding in custom applications	Interactive Q&A, writing, analysis	Complex multi-step tasks requiring real-world actions
Autonomy	Zero, only responds when prompted	Reactive, waits for user input each turn	Proactive, plans, decides, and acts independently

Levels of autonomy in AI agents ^[1]



Level	Role	Key capabilities	Human involvement
0	No AI agents	ML models perform predefined tasks; no adaptation	Scientist does everything: hypothesis, tasks, execution
1	Assistants	Simple hypotheses; narrow protocol design; strong reasoning in selected tasks	Scientist defines hypothesis and tasks; AI executes
2	Collaborators	Hypotheses from data trends; rigorous protocols; interprets findings within broader context	Scientist proposes, AI refines; AI plans and executes tasks
3	Scientists	Creative, de novo hypotheses; develops new methods; continuous learning and adaptation	Minimal; AI drives discovery autonomously

AI agents in the scientific research loop



Autonomous iteration...
- from goal to discovery

Continuous refinement...
- output informs next planning cycle

Why not just using raw LLMs directly instead?



Raw LLM	AI agent (better alternative)
Risk of hallucinations	Action & verification
Creative brain, no grounding	Grounded research partner
No built-in fact checking	Executes & reports with sources
“ Garbage in, garbage out ”	Integrates live, validated data

“ An agent facilitates interfacing with LLMs in the biomedical space, while at the same time **safeguarding their functionalities** through sensible constraints! ” [3]

Hallucination in action: an example



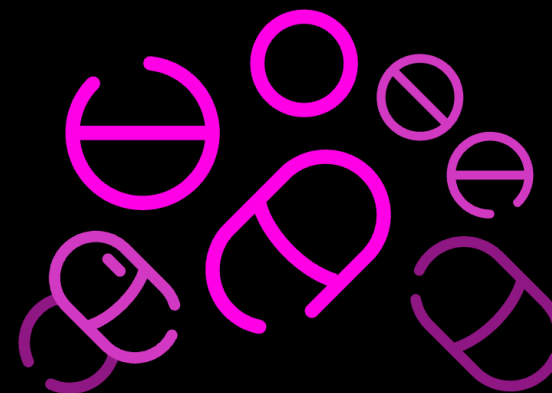
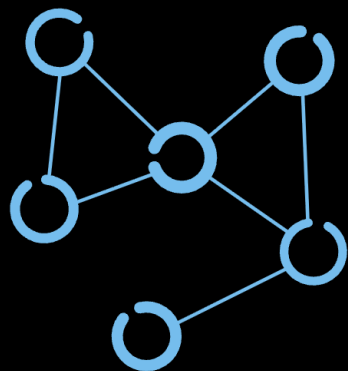
Raw LLM	AI agent (better alternative)
✗ “ The gene BRCA1 is located on chromosome 13 and is primarily involved in DNA repair ... ”	✓ Agent queries the NCBI Gene database → retrieves verified record “ BRCA1 is located on chromosome 17q21.31 (<i>Source: NCBI Gene, ID 672</i>) ”



Knowledge is knowing a
tomato is a fruit.

Wisdom is not putting
it in a fruit salad.

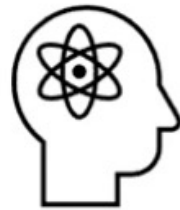
— Miles Kington



How is Life science research being done today?

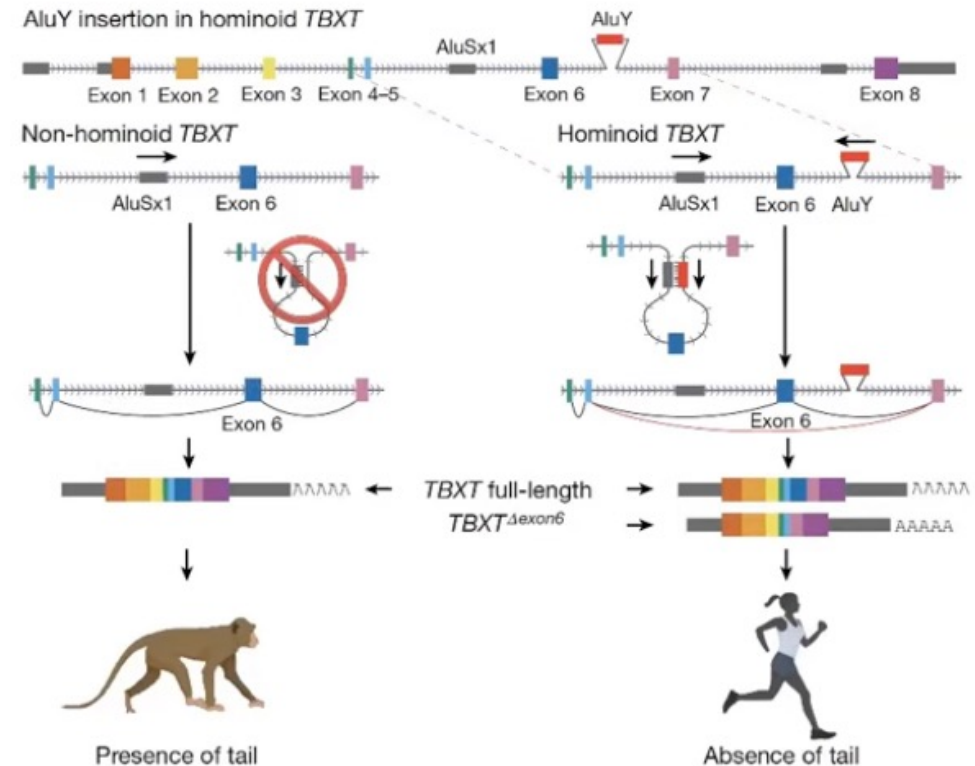


(1) Generating a hypothesis



biologist

“TBXT causes tail loss.”



How is Life science research being done today?



(2) Validating a hypothesis



Biostatistician

Reasoning why a variant could cause a phenotype.

Statistical analysis from massive data generates p-values

GTExPortal

biobank^{uk}
Enabling scientific discoveries that improve human health

RCSB PDB
PROTEIN DATA BANK

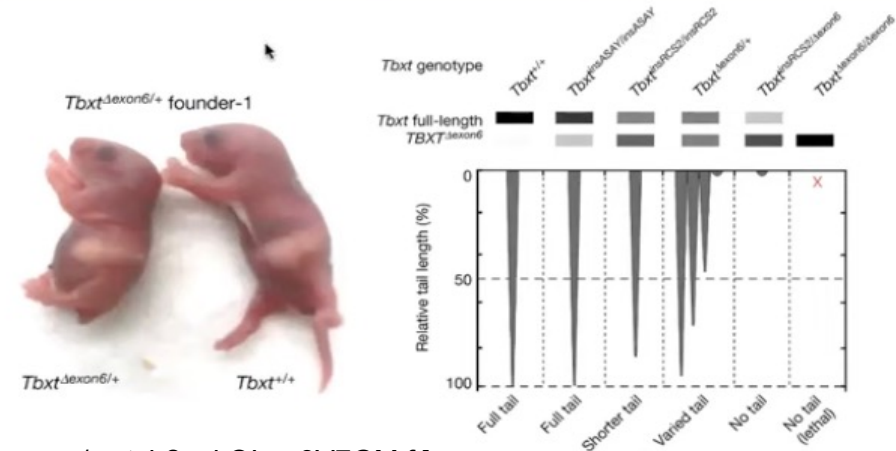
GWAS Catalog



biologist

A high throughput screen to search for genes that regulate a phenotype.

Targeted wet-lab experiment



Life science research at a glance



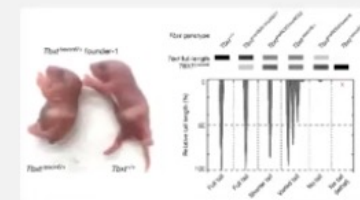
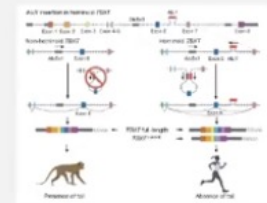
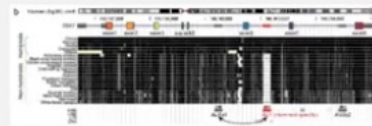
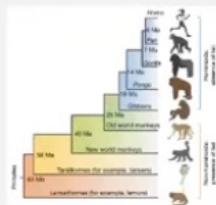
Outer-loop

Hypothesis generation

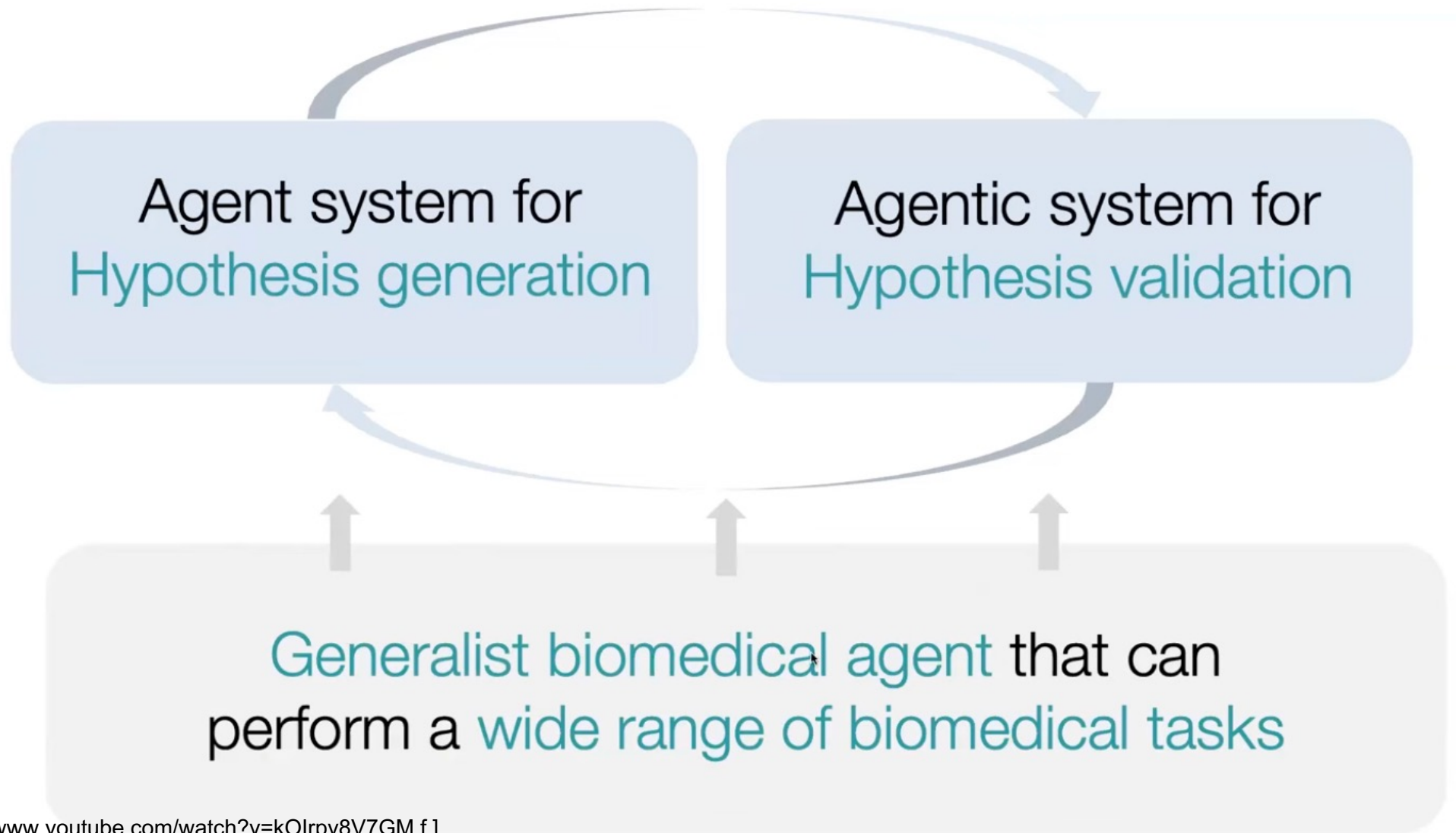
Hypothesis validation

Inner-loop

Massive number of individual workflows



A framework for building collaborative AI agents in life science



In the quest for a life science agent...

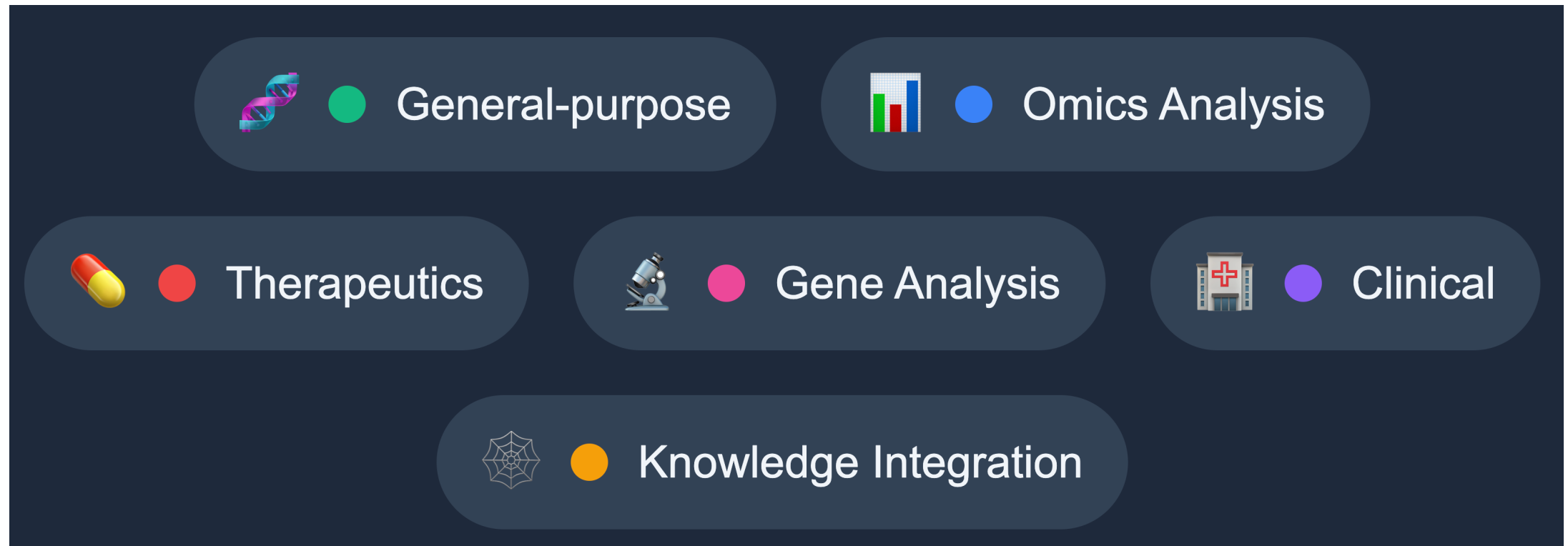


Capability	Details
Adapt	Plug in tools from genomics, drug discovery, clinical data, ... any domain
Unify	Orchestrate workflows across multiple domains
Evolve	Add new skills as research advances
Efficient	Automate repetitive data tasks
Reproducible	Transparent, executable record of every step
Scalable	Tackle complex, multi-step research at scale

The AI agent landscape in life sciences (2024-2025)



Let's explore the key developments of the past two years...



[5]

Finding the right agent for the job



If you need to...	Consider...
Analyze RNA-seq/scRNA-seq data	AutoBA, CellAgent
Explore spatial transcriptomics	SpatialAgent
Understand gene functions	GeneAgent
Design CRISPR experiments	BioDiscoveryAgent
Research drug interactions	TxAgent
Query biomedical knowledge graphs	BioChatter, ESCARGOT
General biomedical research tasks	BioChatter, Biomni
Clinical/medical decision making	MDAgents

Getting started: no-code web-based tools



BioChatter*



CellAgent



Biomni*



GeneAgent

*Biomni is also available via the **SciLifeLab Serve** platform.
Contact: serve@scilifelab.se

Dive deeper with ***BioChatter** : SciLifeLab AI seminar series



Speaker: Sebastian Lobentanzer

- Principal Investigator, Helmholtz Center Munich
- Affiliated with Open Targets, EMBL-EBI
- Head of Computational Biology, German Center for Diabetes Research

Join the Discussion (No registration needed!)

- **When:** Friday, February 13, 10:00 - 11:00 CET
- **Where:** Zoom, meet.nbis.se/aiaio



Event page



Zoom Link



BioChatter Tool

Title: Agentic AI in the life sciences – toward trustworthy research environments

What you'll learn

- How benchmark-first development addresses reliability and hallucination risks
- Architecture for reproducible workflows (moving to Integrated Research Environments)
- Practical implementation of transparent tool orchestration

Privacy, ethics & responsible use



Your data type	Recommended approach	Why it matters
Human patient or clinical data	Install locally or use institutional servers	Keeps sensitive information within your firewall and compliant with institutional policies
Published or synthetic datasets	Web tools are fine	Public data carries no privacy risk
Collaborative multi-site studies	Check with your data access committee first	Ensures all institutions agree on AI tool usage

Legal Aspects of AI Agents in Life Sciences



- **EU AI Act:** Healthcare AI is high-risk; requires transparency, human oversight, and conformity assessments
- **Data protection:** GDPR applies; ensure consent and caution with data leaving the EU
- **Liability:** Responsibility for AI errors; document AI decisions for accountability
- **Sector compliance:** Clinical/diagnostic AI may require medical device regulation (MDR/IVDR)

SciLifeLab AI Strategy



- **Coming Soon!**
- **Currently in development:** Draft compiled in 2025, discussed at the AI hearing (October 2025)
- **Final version being compiled now:** Official strategy release expected soon
- **Sign up** to be notified when the strategy is available →



Good practices for publication



- **Disclosure:** Journals appreciate transparency, mention “analysis pipeline designed with [Agent Name]” in methods
- **Reproducibility:** Save the agent's execution logs (most tools provide these) so others can follow your steps
- **Diversity awareness:** Many biomedical databases have ancestry biases, interpret results carefully if working with underrepresented populations

nature

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[nature](#) > [news explainer](#) > article

NEWS EXPLAINER | 03 October 2025 | Correction [07 October 2025](#)

How AI agents will change research: a scientist's guide

Researchers are increasingly turning to artificial-intelligence tools that can handle complex, multi-step processes.

By [Elizabeth Gibney](#)

[6: Nature: <https://www.nature.com/articles/d41586-025-03246-7>]

Setting realistic expectations



- Agents are powerful partners, but not perfect (yet 😊)
- Common situations to navigate,

Situation	What to keep in mind
The agent suggests an unexpected analysis step	Ask <i>why</i> , good agents explain their reasoning; great ones show their sources
Results differ from your lab's previous findings	Agents query public databases that update regularly, check version dates
The output looks "too confident"	Your intuition is still the best quality control, Do not trust, always verify

Simple safeguards



- ✓ **Start with what you know:** Test the agent with a gene or pathway you understand well before exploring new territory
- ✓ **Check the timestamp:** Database queries reflect the last update cycle, verify if that fits your research timeline
- ✓ **Keep the human in the loop:** The agent handles execution; you handle interpretation

“ Think of it as a skilled research assistant; speeds up the workflow, but you still review the final draft before submission. ”

The five-minute comfort test



Before using any agent on your real research data:

1. Ask it something you already know (e.g., “What does BRCA1 do?”)
2. Check if the answer matches your knowledge and includes source links
3. If yes  → you're ready to try your actual research question

Emerging trends



Trend	Examples	Impact
Self-verification	GeneAgent, ESCARGOT	Reduces hallucinations in scientific outputs
Multi-agent collaboration	MDAgents BioChatter	Simulates expert team consultations
Tool-augmented reasoning	TxAgent Biomni	Grounds AI in validated databases
Code generation + execution	AutoBA, CellAgent, Biomni	Enables end-to-end analysis pipelines
Knowledge graph integration	ESCARGOT, BioChatter	Structures biomedical relationships

What's next



- **Expanding tool ecosystems:** More databases, more analysis methods integrated
- **Wet-lab integration:** Agents that can design AND help execute experiments
- **Regulatory considerations:** As agents enter clinical settings, validation becomes crucial
- **Community building:** BioContextAI, open-source contributions
- **Multi-modal agents:** Combining text, images, molecular structures

What's next



- **Unprecedented speed:** Automate entire workflows, compressing weeks of work into hours.
- **Parallel exploration:** Tackle multiple complex questions and experimental pathways *simultaneously*.
- **Augmented creativity:** Break through cognitive barriers by exploring novel, data-driven hypotheses you might never have considered.

Concluding remarks: embracing the evolution



Where we stand today...

✗ AI agents are imperfect → they will frustrate you

✓ But progress is rapid: impossible → routine in six months

Concluding remarks: embracing the evolution



The researcher mindset...

- ✓ Stay skeptical, but be cautiously curious
- ✓ Early adopters can influence how these tools and their safeguards evolve.
- ✓ Better to guide the evolution than catch up later

Concluding remarks: embracing the evolution



Science has always adapted and evolved...

- ✓ Sanger sequencing → CRISPR
- ✓ Paper notebooks → electronic databases
- ✓ Adaptation is our core principle; change is our constant

Concluding remarks: embracing the evolution



Moving forward...

- ✅ Start small, test carefully, stay critical
- ✅ Don't step away from the conversation
- ✅ **The future belongs to those who adapt!** 😊

References: selected AI agents (alphabetical)



- i. AutoBA: <https://github.com/JoshuaChou2018/AutoBA>
- ii. biochatter: <https://github.com/biocypher/biochatter>
- iii. BioDiscoveryAgent: <https://github.com/snap-stanford/BioDiscoveryAgent>
- iv. Biomni: <https://github.com/snap-stanford/biomni>
- v. CellAgent: <https://arxiv.org/abs/2407.09811>
- vi. ESCARGOT: <https://github.com/EpistasisLab/escargot>
- vii. GeneAgent: <https://github.com/ncbi-nlp/GeneAgent>
- viii. MDAgents: <https://github.com/mitmedialab/MDAgents>
- ix. SpatialAgent: <https://github.com/Genentech/SpatialAgent>
- x. TxAgent: <https://github.com/mims-harvard/TxAgent>

References:



- [1] Cell: <https://www.sciencedirect.com/science/article/pii/S0092867424010705>
- [2] AI-generated by Mahbub using ChatGPT (5.2 Pro)
- [3] BioChatter slides: https://ftp.ebi.ac.uk/pub/training/2025/Webinars/20250310_LLM-driven_bioscience%e2%80%9494PDF.pdf
- [4] Biomni talk: <https://www.youtube.com/watch?v=kOIrpy8V7GM>
- [5] AI agent landscape tool by Mahbub: <https://shorturl.at/MOY7d>
- [6] Nature: <https://www.nature.com/articles/d41586-025-03246-7>

Thank you! 😊



Contact: datacentre@scilifelab.se, serve@scilifelab.se

Explore the AI agent landscape tool 😊

Use the link below, or scan this QR code

<https://shorturl.at/MOY7d>



The future belongs to those who adapt! 😊